

Lem aprox indicatriz compacto abierto continua

Lema 1. Sea $V \subset \mathbb{R}$ un abierto y $K \subset \mathbb{R}$ un compacto tales que $K \subset V$

$$\implies \exists g \in \mathcal{C}_c(\mathbb{R}) : \forall x \in \mathbb{R} : \mathbb{1}_K(x) \leq g(x) \leq \mathbb{1}_V(x).$$

Demostración: Como V es abierto, $\exists \{I_j\}_{j \in \mathbb{N}}$ intervalos abiertos tales que $V = \bigcup_{j \in \mathbb{N}} I_j$. Como K es compacto y $K \subset V$ (luego $\{I_j\}_{j \in \mathbb{N}}$ es un cubrimiento por abiertos de K), $\exists \{I_{j_\ell}\}_{\ell=1}^m$ tal que $K \subset \bigcup_{\ell=1}^m I_{j_\ell}$.

Definimos $\forall \ell \in \mathbb{N}_m : K_\ell := K \cap \overline{I_{j_\ell}}$, entonces K_ℓ es un intervalo compacto y $K_\ell \subset I_{j_\ell}$. Por el `lem-aprox-indicatriz-continua/??`, tenemos que

$$\forall \ell \in \mathbb{N}_m : \exists g_\ell \in \mathcal{C}_c(\mathbb{R}) : \forall x \in \mathbb{R} : \mathbb{1}_{K_\ell}(x) \leq g_\ell(x) \leq \mathbb{1}_{I_{j_\ell}}(x).$$

Ahora definimos $g = \sum_{\ell=1}^m g_\ell \in \mathcal{C}_c(\mathbb{R})$, entonces

$$\forall x \in \mathbb{R} : \mathbb{1}_K(x) \leq \sum_{\ell=1}^m \mathbb{1}_{K_\ell}(x) \leq \sum_{\ell=1}^m g_\ell(x) = g(x) \leq \sum_{\ell=1}^m \mathbb{1}_{I_{j_\ell}}(x) \leq \mathbb{1}_V(x). \quad \blacksquare$$

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